NAYAN

COMPUTER SCIENCE UNDERGRADUATE | SOFTWARE DEVELOPMENT | AI/ML | CYBERSECURITY

Chennai, Tamil Nadu, India | +91 91427 21695 | nayankumar9987@gmail.com

linkedin.com/in/nayan-nayan-69069233b | github.com/NAYANBECS24

PROFESSIONAL PROFILE

Results-oriented B.E. Computer Science & Engineering (Cyber Security) undergraduate, published patent-application co-inventor, web developer, and technical team lead with experience in software development, applied AI/ML, generative AI, cybersecurity, privacy engineering, edge systems, blockchain, and research. Has designed and deployed end-to-end web applications, contributed to AI internships, led cross-disciplinary hackathon projects, and represented developer and innovation communities. Recognitions include a Top-10 finalist placement at the Indian Army Terrier Cyber Quest 2026, the Young Tech Visionary Award, first place at the NIT Tiruchirappalli AI For ALL Ideathon, and a top-112 Asia finish at the Cardano Hackathon.

EDUCATION

B.E. Computer Science & Engineering (Cyber Security) | Aarupadai Veedu Institute of Technology (AVIT), Chennai 2024-2028

Class XII (CBSE), St. John's Academy, Mahua, Bihar | Class X (CBSE), B.D. Public School, Hajipur (Panapur Langa), Bihar

PATENT / INTELLECTUAL PROPERTY

Adaptive AI-Lightweight Crypto Framework (ALCF) for Real-Time IoT Security | Co-Inventor Published Dec 2025

- Indian Patent Application No. 202541117496 A, filed November 2025; published in The Patent Office Journal 51/2025.
- Three-layer Device/Edge/Cloud IoT security architecture combining ASCON lightweight cryptography and TinyML on ESP32; original prototype claims/targets: <2-second detection, 95% accuracy, 30-50% energy savings, and <200 microsecond encryption.

PROFESSIONAL, RESEARCH & LEADERSHIP EXPERIENCE

Summer Research Intern (Selected) | TATA Centre for AI & ML, NIT Tiruchirappalli May-Jul 2026

- Selected for a six-week Education Track research program following first place at the AI For ALL Ideathon; led Team PARTH on SUTRADHAR under faculty mentorship.
- Developed a contextual generative-AI tutor to localize NCERT material into regional dialects and culturally relevant learning examples.

AI/ML Intern | iTech (AVIT Hackathon 2026 Award) 2026-Present

- Awarded an AI/ML internship after winning AVIT Hackathon 2026, held 15 July 2026 by AVIT in partnership with C3 as an iTech initiative.

Agentic AI Intern | Brain O Vision Solutions India Pvt. Ltd. Jan 2026-Present

- Developing agentic-AI workflows for process automation and decision support in organizational settings.

R&D Intern, Campus Catalyst Program (CCIP 2026) | Brain O Vision Solutions India Pvt. Ltd. Feb 2026-Present

- Contributing to applied AI research and automation in the R&D business division (Intern ID: BOV-CCIP-002).

Web Developer | Eduooze, School of Banking 2025-Present

- Developed the candidate enrollment, placement, and product-sales portal using HTML, CSS, JavaScript, PHP, and MySQL.
- Worked on user interface design, performance, and web deployment. Websites: eduooze.in; eduoozeschoolofbanking.infinityfreeapp.com.

Vice President | CYFENCE Cybersecurity Club, AVIT 2026-Present

- Lead technical events, security awareness sessions, workshops, and faculty-student collaboration for the cybersecurity community.

Student Ambassador | Aarupadai Veedu Institute of Technology 2025-2026

- Represented AVIT in academic and innovation programs; recognized for contribution during Annual Day 2026.

Google Ambassador | Google Student & Developer Programs 2025-Present

- Promote developer technologies and peer learning across cloud, AI, web, and emerging technologies.

Innovation Ambassador (Advanced & Foundation) | MIC & AICTE 2025

- Completed both training levels to promote entrepreneurship, problem-solving, design thinking, and innovation culture.

TECHNICAL EXPERTISE

Programming & Web: Python, PHP, JavaScript, SQL, HTML, CSS

AI, Machine Learning & Deep Learning: TensorFlow, Keras, PyTorch, VGG16, Faster R-CNN, Federated Learning, retrieval-augmented generation (RAG), LangChain, Llama-3, GPT-4o

Generative, Quantum & Edge AI: Stable Diffusion XL, LoRA, Wav2Lip, Tortoise TTS, TensorFlow Quantum, PennyLane, TinyML, agentic AI

Application Development: Flask, FastAPI, React, Tailwind CSS, OpenCV, Tkinter, Streamlit, Jupyter, VS Code
Cybersecurity & Privacy: TOR network analysis, cyber forensics, anonymous-network intelligence, IoT security, ASCON lightweight cryptography, data redaction, blockchain audit trails
Cloud, Tools & Platforms: Google Cloud Platform, Intel oneAPI, OpenVINO, Hostinger, web hosting, Google AdSense
Data & Retrieval: MySQL, MongoDB, Neo4j, Pinecone
Blockchain & Web3: Cardano/Plutus, CIP-68, Atala PRISM, Halo2 zk-SNARKs, Circom, Hydra, Polygon; project exposure to Midnight, Masumi, Milkomeda, Wormhole
Professional: Analytical problem-solving, team leadership, collaboration, rapid prototyping, public speaking, technical presentations, design thinking, research communication

SELECTED PROJECTS & PRODUCT PORTFOLIO

Project SUTRADHAR | Context-Aware Generative AI Tutor for Bharat | *GenAI / EdTech / NIT Tiruchirappalli*

- Led Team PARTH; built the Gurukul Context Engine to adapt NCERT examples into local analogies using Llama-3 and the Bhashini API.
- Designed regional-dialect, lip-synced tutoring with Stable Diffusion XL/LoRA, Wav2Lip, and Tortoise TTS; supported low-bandwidth WhatsApp/SMS learning on 2G/3G. Related recognition: AI For ALL Ideathon first place.

AEGIS | Adaptive Enterprise Guardrail & Intelligence System | *GenAI Security / ET GenAI Hackathon*

- Designed GhostGuard, a shift-left semantic firewall against package-name slopsquatting using an NPM/PyPI knowledge graph, and Chimera, a runtime adaptive-deception/digital-twin engine.
- Technology: Llama-3-8B, GPT-4o, LangChain, FastAPI, Neo4j, Pinecone; aimed at DevSecOps, runtime defense, and threat intelligence.

Q-Shield | Hybrid Quantum-Classical Ransomware Precognition | *Cybersecurity / IEEE Hackathon*

- Proposed pre-encryption ransomware detection from system-call rhythm: C++/Rust kernel agent to graph features, parametrized quantum circuits, and a classical neural classifier.
- Technology: TensorFlow Quantum, PennyLane, TensorFlow, Streamlit.

SOVARTH / Sovereign Nexus | Privacy-First Cardano DeFi for Asia | *Blockchain / Cardano Hackathon Asia 2025*

- Led Team PARTH in designing an identity, asset, credit, and cross-chain wallet concept for underserved Asian users; ranked among the top 112 teams in Asia.
- Used CIP-68 Data-Bound Assets, Charli3 IoT oracles, Midnight/Halo2 zk-SNARK credit scoring, Atala PRISM DIDs, Masumi AI solvers, and Hydra/Milkomeda/Wormhole interoperability. Original design projected 95% lower cost and sub-30-second settlement.

TOR Sentinel | Privacy-Preserving Federated AI for TOR Traffic Analysis | *Cybersecurity / Federated Learning*

- Designed federated analysis across ISPs without sharing raw personally identifiable data, using Adaptive Time-Window Correlation (ATWC) to address variable TOR latency.
- Reported >80% probabilistic tracing confidence in simulation, with blockchain-based judicial evidence trails and simulated investigation times reduced from weeks to hours. Associated with Innoventia 2K26 and the TN Police Cyber Crime Hackathon.

GridWeaver | TCP/IP for Electricity | *Energy Systems / Edge AI*

- Designed an ESP32-S3 peer-to-peer mesh grid with Dynamic Islanding and reported <200 ms blackout recovery in hardware-in-the-loop validation.
- Incorporated LSTM forecasting, ATECC608A hardware security, and Proof-of-Energy-Transfer blockchain.

EchoShield | AI Voice Deepfake & Vishing Defense | *Audio Security / Edge AI*

- Developed a privacy-first, on-device voice-liveness detection approach using breathing, micro-jitters, and room-tone signals; reported 98.7% accuracy and <1.2 s response time.
- Designed for language-agnostic operation, including English, Hindi, Tamil, and Telugu.

SurakshaAI | Privacy-First Road Safety Platform | *Civic Tech / National Road Safety Hackathon 2026*

- Built a design combining predictive accident risk, an offline legal co-pilot, LENS-Net sign detection (reported 92.56% mAP), and encrypted rolling-buffer evidence.
- Applied Circom zk-SNARK proofs to contest incorrect traffic challans without revealing detailed location.

AgriSmart 2.0 / AI Farming 2.0 | *Agriculture / Sustainability*

- Designed AI-assisted crop analysis, automated agricultural decisions, yield optimization, and direct market access.
- Received first prize at the National Science Day 2026 model competition; an earlier AI Farming 2.0 version received third prize at an SRM hackathon.

Result Analyzer | An FA's Pocketeer | *Full-Stack Product / Education Analytics*

- Independently developed an academic-results application with Admin/Faculty/Student portals, bulk Excel uploads, automated SGPA/CGPA calculations, grading analytics, and university-aligned workflows.
- Champion, AVIT Mini Product Showcase 2026; AVIT CSE Department Certificate of Appreciation for successfully implementing the Result Analysis Generator App (signed by the HoD and Principal). Live: resultanalyzer.infinityfreeapp.com.

ID Masker Tool | AI-Powered PII Detection & Redaction | *Privacy Engineering / Computer Vision*

- Trained a Faster R-CNN (ResNet-50) image model to detect sensitive information; implemented black-box, Gaussian blur, and pixelation redaction.
- Created an on-premise privacy-focused interface with PyTorch, OpenCV, Flask, and Tkinter to support GDPR-aligned use cases.

BharatRealEstateAI | AI Real Estate Discovery | ***Applied AI / Intel Unnati Program***

- Designed emotion-aware RAG property discovery with semantic search in 12+ Indian languages, Vastu and joint-family cultural preferences, and visual similarity matching.
- Reported approximately 94% simulated temporal price-prediction accuracy and 180 ms search over 2.5M+ properties using Intel oneAPI/OpenVINO.

NutriLens | AI-First Food & Health Coaching | ***HealthTech / AMD Slingshot Hackathon***

- Architected an AI nutrition/coaching application incorporating Indian regional cuisine and 8+ Google Cloud services.
- Shortlisted for the AMD Slingshot Regional Ideathon in Bengaluru; project leaderboard score: 95.24.

Sanjivani | Decentralized Offline-First Health Ecosystem | ***HealthTech / Pysellers Innovation Challenge***

- Designed an offline-first health ecosystem with quantized Phi-3/TinyLlama edge AI, multi-agent triage, patient-owned DID/AES-256 records, federated learning, Wi-Fi Aware mesh networking, and a Health Points economy.

BioForge Dustbin | AI Smart Waste Segregation & Biogas | ***Smart Cities / Sustainability***

- Designed onboard-AI separation of biodegradable and non-biodegradable waste with local conversion of organic waste into biogas.

JobDham.in | Full-Stack Job Portal | ***Web Development / Live Product***

- Built and hosted a job platform from scratch using HTML, CSS, JavaScript, PHP, MySQL, and Hostinger.
- Integrated Google verification and AdSense monetization; operated as an advertising-supported, revenue-generating product. Live: jobdham.in.

Eduooze | School of Banking | ***Web Development / Live EdTech Website***

- Developed a banking EdTech customer platform for candidate enrollment, placements, and digital product sales; optimized front-end UX and deployment.
- Technology: HTML, CSS, JavaScript, PHP, MySQL. Websites: eduooze.in; eduoozeschoolofbanking.infinityfreeapp.com.

APPLIED CYBERSECURITY INVESTIGATIONS

- KC7: The Cyber Detective Game (2026) - Completed simulated security investigations using Kusto Query Language (KQL), OSINT, phishing analysis, email-communication review, endpoint process events, network logs and passive DNS.
- Earned Clout Defender (21 Aug 2026) and Valdorian Times (2 Sep 2026) badges. Traced social-engineering compromise, a politically motivated influence campaign, suspicious communications, data exfiltration indicators and attacker infrastructure in scenario-based labs.
- Also investigated a hip-hop-themed communications case to identify suspicious exchanges and threat patterns; simulations and game investigations, not professional incident-response engagements.

AWARDS, HACKATHONS & DISTINCTIONS

- Top-10 National Finalist - Indian Army Terrier Cyber Quest 2026 (AI Kavach / Track 2), New Delhi (6-8 Oct 2026). Selected among the top-10 teams in the track from a national competition with 8,200+ participating teams; received a Certificate of Appreciation from the Director General, Territorial Army, Integrated Headquarters of Ministry of Defence (Army), for active participation and exemplary performance. Competed as an AVIT team alongside Divyansh Bharati; faculty mentor: Dr. S. Balakrishnan.
- Winning Team - AVIT Hackathon 2026 (Team PARTH; 15 July 2026), organized by AVIT in partnership with C3 as an iTech initiative. Received a Certificate of Participation; earned an AI/ML internship with iTech following the hackathon win.
- Participant - DECODE SIH 2026, organized by OSCode (Certificate of Participation).
- Participant - QUANT-A-THON 2026, international quantum-technology hackathon at Rajalakshmi Institute of Technology, Chennai.
- Participant - Innovate4FinLit Game Challenge, National Centre for Financial Education (NCFE), powered by Hack2Skill.
- Participant - Mahindra Rise / Breath Beings, The Quiet Power Quiz (#SheIsOnTheRise, 2026).
- Young Tech Visionary Award - Innoventia 2K26, National-Level Project Expo (TOR Sentinel).
- 1st Place - AI For ALL Ideathon, TATA Centre for AI & ML, NIT Tiruchirappalli (February 2026).
- 1st Prize - IdeaForge Ideathon 2026, AVIT (Team PARTH lead).
- 1st Prize - National Science Day 2026 model competition (AgriSmart 2.0).
- Champion - AVIT Mini Product Showcase 2026 (Result Analyzer); Certificate of Appreciation from AVIT CSE Department for implementation of the Result Analysis Generator App.
- 2nd Place and Top 10 Finalist - Tamil Nadu Police Cyber Crime Hackathon 2025.
- Top 112 Teams (Asia) - Cardano Hackathon Asia 2025 (SOVARTH).
- Semi-Finalist - ET-AI Hackathon 2026, The Economic Times (certificate dated 7 May 2026).
- Top 11 Finalist - GIEC Peace Innovation Welfare Challenge.
- Shortlisted - AMD Slingshot Regional Ideathon, Bengaluru (Food & Health App); project submission leaderboard score of 95.24 (rank not provided).
- 2nd Rank - IdeaJam 2025 (national level).
- 3rd Prize - Innovation Showcase 2K26 (INR 500 award).
- 3rd Prize - SRM Hackathon (AI Farming 2.0).
- Jury's Special Appreciation Award - INTELLECTION 2025.
- Top 10 Innovation Champions - AVIT-CSE (December 2025).
- Institution Innovation Competition 2026 - certificate and INR 750 appreciation.

- Qualifier - The Big Code 2026 by Google.
- Selected - Beauty Tech Hackathon 2026.
- Participant - IEEE Hackathon (Q-Shield).
- Participant - ET GenAI Hackathon 2026 (AEGIS).
- Participant - India AI Impact Buildathon 2026 (40,000+ participants reported for the event).
- Participant - Technoverse 2026 Grand Finale, St. Joseph's College of Engineering (14 March 2026).
- Successful prototype submission - Google Solution Challenge 2026 (Build with AI), certificate dated 22 July 2026.
- Consolation Prize - Silver Spark 2025 Ideathon.
- Product Expo Appreciation - IFET College of Engineering.
- 1st Prize - Science Project.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Innovation Ambassador - Advanced Level, Ministry of Education Innovation Cell (MIC) & AICTE (2025).
- Innovation Ambassador - Foundation Level, MIC & AICTE (2025).
- Professional Advancement Program: "Taking IPR (Invention) to Market and to Commerce: Role of IPR" - Vinayaka Mission's Research Foundation with G-CEIP (15 June 2026).
- International-Level Bootcamp on DSA for DeepTech - one-month virtual training, Brain O Vision Solutions India Pvt. Ltd., in association with AICTE and hosted by AVIT/VMRF (1-30 May 2026).
- Student Innovators Boot Camp - Vinayaka Mission's Research Foundation, Salem (6 Aug 2026); problem framing, collaboration and innovation practice.
- Full Stack Software Development, 20-hour value-added course (VA223) - Vinayaka Mission's Research Foundation (2 Aug-18 Oct 2025).
- Design-to-Code with Antigravity and Stitch MCP - Google Developer Program (also listed as "RF Design-to-Code with Antigravity & Stitch MCP").
- From Relational Model (SQL) to MongoDB Document Model - MongoDB (also listed as "SQL to MongoDB Document Model").
- Netrix 2025 Certification.
- Google Ambassador - Google Student & Developer Programs (community role/recognition).

PROFESSIONAL STRENGTHS & INTERESTS

Strengths: Team leadership, product ownership, ethical AI, collaborative engineering, research-oriented problem solving, stakeholder communication, rapid prototyping and technical presentation.

Interests: Cybersecurity research, responsible AI, quantum computing, blockchain/Web3, decentralized systems, sustainable technology, innovation programs and hackathons.

Career interests: Internships, software engineering, AI/ML engineering, cybersecurity, full-stack development, R&D, innovation and technology/product roles.